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TOWARDS ECO-SOCIAL CONTRACTS THROUGH AN ENTANGLED MANIFESTATION OF DIGITAL LITTER

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ABSTRACT

Many of the current debates in technology concern the extractive practices around planetary resources and data, which produce electronic waste as well as everyday sense of clutter among media channels and files. In this paper, we engage with this problem space through a specific focus on Terms of Service agreements, which have become a prerequisite for even the most rudimentary use of technology. We do this by presenting an exploded view of an existing Terms of Service policy ecosystem, in the form of an outdoor installation, temporarily *littering* a patch of nature. Through this installation, we expose *eco-social contracts* as a design opening for more-than-human data governance, *ecological temporalities*, and *the right to patchwork* as generative concepts.



Figure 1. Overview image of the installation.

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INTRODUCTION

"Modernity is a condition of compulsive, and addictive, designing. Where there is design, there is waste." says Zygmunt Bauman (2004, p.30). Indeed, if we compare our lives with previous generations', a dramatic increase in the amount of daily trash produced could be observed. This is true also in the context of computational media, in the form of 'infotrash' (Offenhuber, 2025) that occupy websites, platforms – "online services of content intermediaries" (Gillespie 2010, p.348), and inboxes, a form of waste that often piles up away from our awareness. In this paper, we use the notion of *litter* as a critical lens for attending to computational waste, which we present through a material manifestation of Terms of Service agreements, enacted in a forest in Umeå, Sweden (Figure 1) (Özçetin, 2023b; Özçetin & Özçetin, 2023). By presenting our reading of this manifestation, we unearth problematic aspects of how we are made to relate to digital technologies. We use these insights for redirections.

In what follows, we start with introducing related work. We then detail the installation as a sense-making exercise and follow with a reflection through an analysis of *littering* practices performed through Terms of Service agreements. We then discuss *eco-social contracts* as a design opening towards democratic data governance and, *ecological temporalities* and *the right to patchwork distributed computational things* as relevant concepts.

RELATED WORK

Terms of Service agreements and accompanying Privacy Policies are online phenomenon commonly referred to as everyday annoyances. Several studies report low engagement with and comprehension of these (Bakos et. al, 2014; Maronick, 2014; Obar & Oeldorf-Hirsch, 2018). In response, from a perspective of redesign at the interface level of 2D screen interfaces (Özçetin & Wiltse, 2023), various studies explored visual design principles (Kelley et al, 2009; Kay & Terry, 2010; Tabassum et al, 2018; Taber et al, 2020; Reinhardt et al., 2021; Habib et al, 2021; Kitkowska et

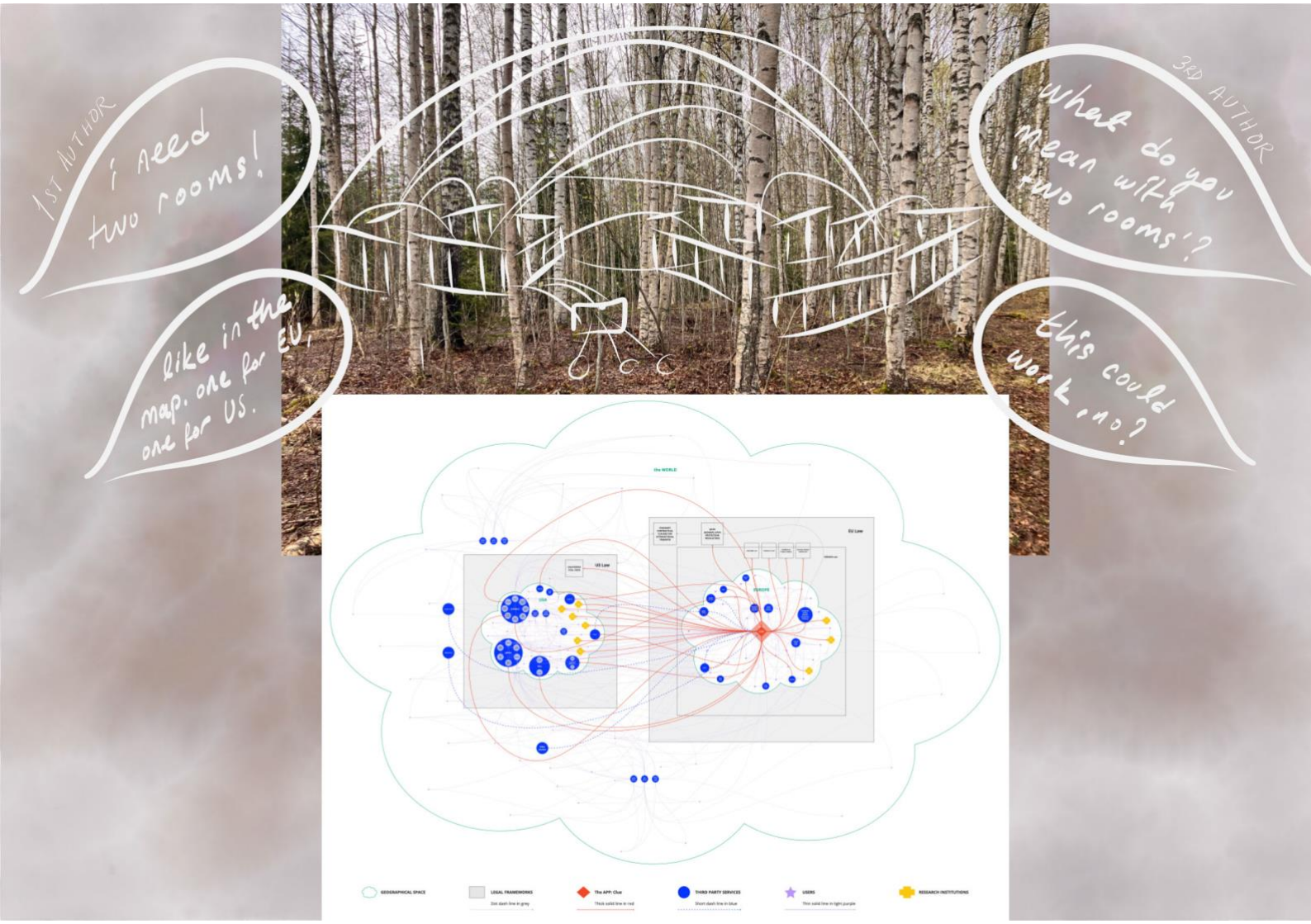


Figure 2. Using the 2D *relational and power map* (Özçetin, 2023a; Özçetin and Wiltse, 2023) as reference to find a suitable location

al., 2022), modularity (Das et al, 2018), data packages (Jones et al., 2017), and simplified language (Robinson & Zhu, 2020).

Other design projects have explored how to make visible what is consented to (Nicenboim, 2015; Pierce et al., 2018; Jansson; 2019; Özçetin & Wiltse, 2023). Drawing attention to the hidden entanglements in such agreements through *relational and power maps* (Figure 2), Özçetin and Wiltse (2023) proposed seeing these as relational sites rather than as 2D interfaces. This understanding proposes a conceptual shift from singular Terms of Service agreements, towards '*ToSspheres*' – dynamic and distributed ecosystems of policies from the primary service, its third-party services, and the binding legal frameworks (Özçetin, 2023a; Özçetin & Wiltse, 2023; Özçetin & Redström 2024b). Such translation of texts into maps surfaces the internal relationality, making interventions possible (Özçetin & Özçetin, 2023; Özçetin & Wiltse, 2023), yet still provides a flat and isolated picture.

The commonality among the above approaches is that they make visible a situation that has become familiar through repetitive encounters. The Russian formalist art critic Victor Shklovsky (1917, p.11) explains this as 'algebrization' of perception: "we know what it is by its configuration, but we see only its silhouette. The object, perceived thus …, fades and does not leave even a first impression; ultimately even the essence of what it was is forgotten.". It is through *defamiliarization*, by making familiar things unfamiliar, Shklovsky (1917, p.12) argues that perception can be delayed, and an object is experienced "in its continuity" rather than "its extension in space". He exemplified this through literary texts, but recently *defamiliarization* has also been discussed in design research, for instance in relation to ethnographic accounts (Bell et al., 2005), embodied ideation methods (Wilde et al., 2017), and counterfactual artifacts (Hendriks et al., 2024).

Similarly, this project is drawing on *defamiliarization* as a sense-making tool to enable a prolonged engagement with the *ToSsphere* as a relational site where interactions take place to explore what becomes visible when messiness of and pluralities in complexity is favoured over the static representations in system maps that Josina Vink (2023) critiques.

A *TOSSPHERE* IN A FOREST

The *defamiliarization* exercise took the form of a temporary performative installation in an urban forest in Umeå, Sweden, materializing an 'exploded view' of *Clue*[1] application's *ToSsphere*, as illustrated in a previously designed *relational and power map* showing the weight of different entities with a planetary distribution (Özçetin, 2023a; Özçetin & Wiltse, 2023) (Figure 2). At a glance, the installation presents an overwhelming amount of papers with policy agreements printed on both sides, hung around and between trees, connected to each other with threads. The activity was led, planned and performed[2] by the first author, with the third author documenting the process with photographs and video clips showing incremental improvements in



Figure 3. Materials used for the mapping such as policies, reference map, ribbon, cotton balls, paper tape, paper clips



Figure 4. A screenshot of visual documentation library of the mapping showing a small number of video clip thumbnails

1 *Clue* is an application for reproductive cycles: https://helloclue.com

2 For the mapping performance, the first author dressed in white shirt and off-white trousers in linen, beige trench coat, yellow rainboots, and a headscarf from her country of origin to speculate on a look of a *ToSsphere* spatial mapper that challenges stereotype of policymakers as 'men in blue suits'.

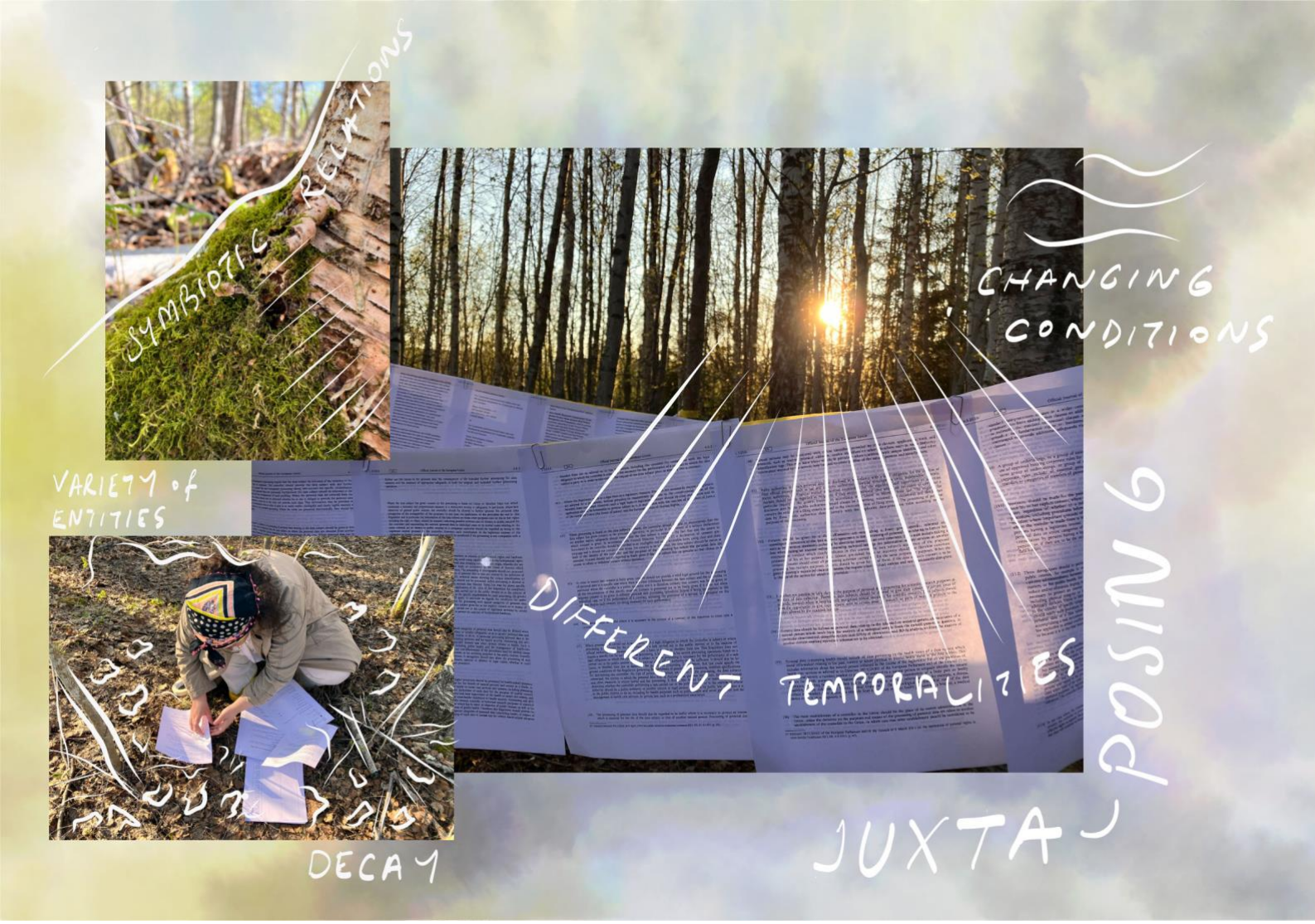


Figure 5. Juxtaposing the *ToSsphere* to the forest

the mapping, close-ups, environmental context and the finalized mapping (Figure 4).

Below, we introduce this sense-making exercise through key decisions based on working through contrasting concepts by drifting towards what *ToSsphere* was not: from 'immaterial' to 'material' through *materialization*, from 'screens' to 'forest' through *juxtaposition*, and from 'scrollable' to 'strollable' through *spatialization.*

MATERIALIZATION: FROM IMMATERIAL TO MATERIAL

As an ontological and relational approach, data physicalization can deconstruct and defamiliarize a phenomenon (Offenhuber, 2020). Therefore, the first tactic was to materialize the policies of the reference map by printing them double-sided on A4 papers (Figure 3, 6). This made visible the sheer volume of policies and transformed them into a design material allowing different forms of explorations. Loose pages introduced flexibility, enabling them to be freely hung. This decision helped move past the immateriality imposed by the written language presented as scrollable texts within the boundaries of on-screen interfaces.



Figure 6. The first author holding the stack of printed out policies of the *ToSsphere* prior the mapping activity



Figure 7. The first author strolling in the *ToSsphere*

JUXTAPOSITION: FROM SCREENS TO FOREST

Printing the policies decontextualized our *ToSsphere* affording it to be juxtaposed to a space of our choice. Embracing emergence in research through design, to stage an encounter in an unusual setting (Gaver et al., 2022), to get as far as one could get from the screen-bound Terms of Service agreements, going outdoors to a local forest was preferred.

Several hours were spent over two days to scout sites to accommodate the 'two rooms in the reference map', where the dispersion of trees could allow navigating and forming relations between entities. For a clear regional grounding, the location chosen was one mostly occupied with birch trees (Figure 2). The area faced a walking path that facilitated documentation while adding a semi-public element to the activity as strollers glanced at the ongoing mapping[3]. As with data physicalization, space, light, and shadow become additional variables of representation and perception (Segal, 2022), this juxtaposition introduced an unusual context with diverse entities including trees, bugs, rocks, moss, sun, wind, all of which played a unique role in the experience (Figure 5).

SPATIALIZATION: FROM SCROLLABLE TO STROLLABLE

Trees were appropriated as different entities of the 2D map through a mapping that took over two days to complete (Figure 7). This started with assigning two trees to represent the main application, hanging its policies between them. Afterwards, the approach was to select larger trees for third-party service providers heavily present in the reference map, like *Google*, and smaller trees for less present ones, like *Slack*. As trees were selected and labeled, their respective policies were hung around them, while the relations between services were illustrated by pulling threads between the trees. Then, the European and US legal frameworks were hung around two separate outer rows of trees to close this ecosystem. Mirrors were placed across the installation as a metaphor for multiplying the connections and interactions. Finally, cotton balls

[3] We did not monitor what people did when we were away for lunch and between two days. However, after we returned from a lunch break, we noticed that one of the mirrors placed in the installation was stolen.

introduced as ‘users’ were made to pass through an ‘I agree to Terms of Service’ touchpoint (Figure 8).

By manifesting this *ToSsphere* as an inhabitable space (Özçetin & Redström, 2024a) it became strollable rather than scrollable. The embodied experience of this type of spatialization welcomes all senses, movement and a plurality of perspectives (Loke & Robertson, 2013; Wilde et al, 2017). Unchoreographed physical body movement in space, provided new perspectives and revealed new things, compared to scrolling which always gives the same starting point and flow.



Figure 8. ‘Users’ represented as cotton thread balls entering the *ToSsphere* through ‘I agree to Terms of Service’ touchpoint

Additionally, the changing sunlight highlighted different policy sections benefiting from the unbound and distributed arrangement. Together with the visible decay, death, regeneration, and symbiosis in the nature, ecological temporalities and relationalities that they enable became noticeable encouraging a reflection on the disembodied temporality of digital technologies.

Walking on the ground, feeling the moss, tripping over fallen branches and laying rocks provoked contemplating on the myth of the cloud and the materiality of electronic devices and their digital infrastructures: the precious metals in chips, the energy that runs servers, the clean water that cools them down (Crawford, 2021). These dependencies bring light to human societies’ unaccounted dominance over nature suggesting a shift in more-than-human concept in data-intensive contexts from technologies with their own intentionalities (Coulton & Lindley, 2019; Giaccardi & Redström, 2020; Redström & Wiltse, 2019) often entangled with capitalist values of accumulation, to multi-species, whose own intentionalities are often diminished by these same values (Chen et al, 2025).

A PROLONGED ENGAGEMENT: PERCEIVING LITTERING

At the end of the second day, the small patch of the forest was decontaminated, by cutting and unravelling the threads, and collecting all the parts of the installation. The papers were recycled, threads were trashed, clips, mirrors, and balls were saved. Decontaminating the space erased the material presence but gave a lasting personal impact. The prolonged engagement with the *ToSsphere*, through *defamiliarization*, revealed that we were not *‘seeing the forest for the trees’.* Taking the papers printed with Terms of Service texts back to their material source, the trees, created an interesting conceptual contrast. The out-of-place view of hung policies among trees, fallen papers on the ground not standing the wind, the damp pile of policies after the unmapping, the entangled threads, and the act of cleaning after, gave us an impression of *littering* (Figure 9). We collected our litter, yet our minds were littered with the sheer littering that Terms of Service agreements can introduce to our physical and digital lives. Analyzing the work through unstructured reflective dialogue between the first and third authors following the mapping, dialogues over the photographs and edited film (Özçetin, 2023b) and versions of this text led us to reaching out for the literal meanings of *littering* and translating its characteristics of *broken materiality*, *neglected relations,* and *costing acts* into a framework.

BROKEN MATERIALITY

One of the meanings of *litter* refers to human-made materials that have been inappropriately discarded as they were no longer deemed valuable for use. Once we were done with mapping, printed agreements became *litter* as they were damaged for storing to be used again. Similarly, as the everyday use of Terms of Service is reduced to a symbolic gesture of passing through with a click on ‘I agree’ checkboxes, these agreements become *litter*. Until one gets removed from a platform for violation of terms, these mostly remain as ‘single-use’ touchpoints. The virtual waste as excessive amounts of piled up policies to be read and evaluated raises questions as ‘is the information presented in these agreements of use to ‘users’ or is it something *littering* their digital spaces?’, ‘is a bunch of legalese dumped on ‘users’ so that companies cannot be held accountable?’. *Broken materiality* urges us to ‘design away’ (Tonkinwise, 2014; Fry, 2008; Fry, 2005) the current modes of digital agreement with online services.

NEGLECTED RELATIONS

Dumped, abandoned, and left behind as scattered on spaces in an unorganized manner, physical *litter*



Figure 9. Perceiving *littering* throughout the mapping and unmapping the *ToSsphere* in the forest

signifies neglect, carelessness, and a lack of respect for others and shared spaces. It breaks social contracts.

A few meters away from the mapping site there was dumped parts of a vehicle contaminating this peaceful place working against the notion of 'right to roam without causing harm'. Performing the mapping next to this careless act of physical *littering* surfaced how Terms of Service agreements never addressed the environmental harm caused by everyday online engagements. While many relations and dependencies are highlighted in policies, our mapping foregrounded the absence of consideration of nature as an entity in these policies, despite its role in making digital infrastructures and sustaining digital platforms (Crawford & Joler, 2018; Crawford, 2021; Özçetin & Wiltse, 2023). Revealing how nature is made invisible, *neglected relations* urges us to facilitate nature's agency.

COSTING ACTS

Littering points to the scattering things in an unorganized, careless, or overwhelming manner. *Litter* occupies spaces, creating an unnatural and harmful intrusion into ecosystems. Ignoring waste collection systems, *litter* requires additional resources to declutter and decontaminate spaces. *Littering* therefore wastes time and effort. It costs.

Our mapping temporarily cluttered and contaminated the forest. Similarly, current forms of Terms of Service agreements intrude into our everyday, clutter our minds, and waste our time. The time spent for mapping and unmapping this installation hints the unimaginable personal time and effort appropriated by companies if people were to read them to make judgements, as estimated by McDonald and Cranor (2008). The dilemma of choosing between spending time to read unnegotiable terms and contaminating their sense of authenticity one unread policy at a time costs users their agency. *Costing acts* urges us to find more meaningful ways for users to weigh in these contracts.

DESIGN OPENINGS

Examining how digital economy uses waste production, 'infotrash', to generate value, Offenhuber (2025, p.337) argues that "waste is not an incidental by-product of convenience, it is often the purpose that drives entire industries". He suggests looking for the messiness and the materialities hiding behind sleek interfaces to form a material critique.



Figure 10. The right to patchwork one's own distributed computational thing inspired by the act of choosing trees for the mapping

The notions of *littering* that we employ here highlight aesthetical, ethical, and economic dimensions of the 'infotrash' Terms of Service agreements consciously generate, for 'users' and nature, to legitimize business practices. The value in this case comes from dismissing users' and nature's agencies, making their roles invisible or forcing their resignation.

On the other hand, the value of our *littering* act, comes from turning this patch of the forest into a 'pluriversal contact zone', a concept proposed by anthropologist Mariso de la Cadena, where "relational (the emergent story) and nonrelational (the dominant story) narratives of life and worldmaking meet" (Escobar et al., 2024, p.152). By staying with the trouble (Haraway, 2016) of *littering*, we ask how we, the designers of computational things in broader sense, could design with the awareness of such *littering* practices, while looking for ways to prevent them.

Perhaps, embracing the paradoxical nature of design is where we need to start to be conscious of what we include as we exclude, what we enable as we disable, what we set free as we control: "The response-ability that design is, must become a life-affirming, enlivening activity, yet it cannot be conceived as "positive," it has to be understood as paradoxical, pharmacological." (Ávila 2022, p.150) – medicine and poison are both context-dependent. To facilitate consent, Terms of Service creates increasing amounts of *litter* that forces resignation (Seberger et al., 2021) as we no longer can afford to care about the digital and physical spaces such agreements appropriate. As stated through the notion of 'ontological design', the world we design designs us back (Willis, 2006).

Thinking how we can design a world, that acknowledges our interdependencies, we propose a shift towards *eco-social contracts*, "…a natural contract of symbiosis and reciprocity" (Serres 1995, p.38) that includes considerations for the needs of individual users, nature, and businesses. Defining this as 'attunement to one another – human and non-human', here we present possible concepts for design researchers, practicitioners, and policymakers for prototyping design practices for more-than-human data governance.

First, to enable nature's agency, we propose turning toward indigenous knowledges fostering mutual respect through an understanding of nature as a living entity that we are all part of rather than something to be owned and exploited for economic benefit (O'Shea, 2021). Furthermore, taking *ecological temporalities* as our guides, we propose exploring what decay, death, rest, rejuvenation, and ongoing symbiotic relationships could

mean in computational things. We suggest shaping services that take accountability for the environmental contamination and costs caused by the deployment and maintenance of their infrastructures and look for long-term reciprocal relationships with ecosystems - such as the Solar Protocol (2021), a distributed solar-powered servers network, rather than treating environmental impacts as individuals' responsibilities.

Secondly to enable users' agency, we propose shifting our attention from agreements to the figuration of computational things. The experience of choosing trees to represent entities in the policy ecosystem resembles the choices 'users' make when purchasing face-to-face services, where decisions are informed by values as reflected through products, advertisements, price, and so on. What if we took this understanding to computational things that are more like *fluid assemblages* of several entities (Redström & Wiltse, 2019) and proposed 'users' *the right to patchwork* the *distributed computational things* as a way of enabling their agency? (Figure 10). From the perspective of *littering*, just as in 'right to roam without harm', this would empower 'users' to choose entities holding the *eco-social contract*.

Patchworking, making a whole out of pieces, is common in programming, as the practice of altering and reusing snippets of code illustrates. By extending this task to 'entity level', we introduce *interchangeability of entities* building on *interoperability between entities* granting individuals the right to move data across services (European Commission, 2022). Distributing the power to build an ecosystem that now sits with companies would promote more conscious practices across the industry and support more diverse markets by reducing the power of gatekeeping companies. In practice, 'users' could align with companies on values rather than negotiating individual articles with the support of a labelling system of *ToSspheres*.

CONCLUSION

Working further on the relationality in *ToSspheres*, this sense-making exercise through its embodied and spatial qualities shifted our attention to the messiness in the hidden away relations of Terms of Service. *Littering* revealed insights about agency, temporality, and participation, which led to *eco-social contracts* as a design opening into more-than-human data governance and *ecological temporalities* and *the right to patchwork distributed computational things* as related concepts. Moreover, we illustrated how drawing the characteristics of a context at hand and then drifting towards contrasting concepts could enable *defamiliarization*.

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